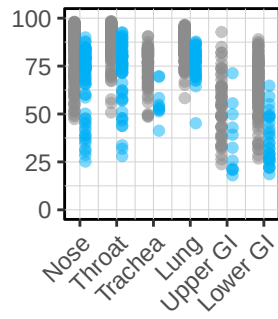
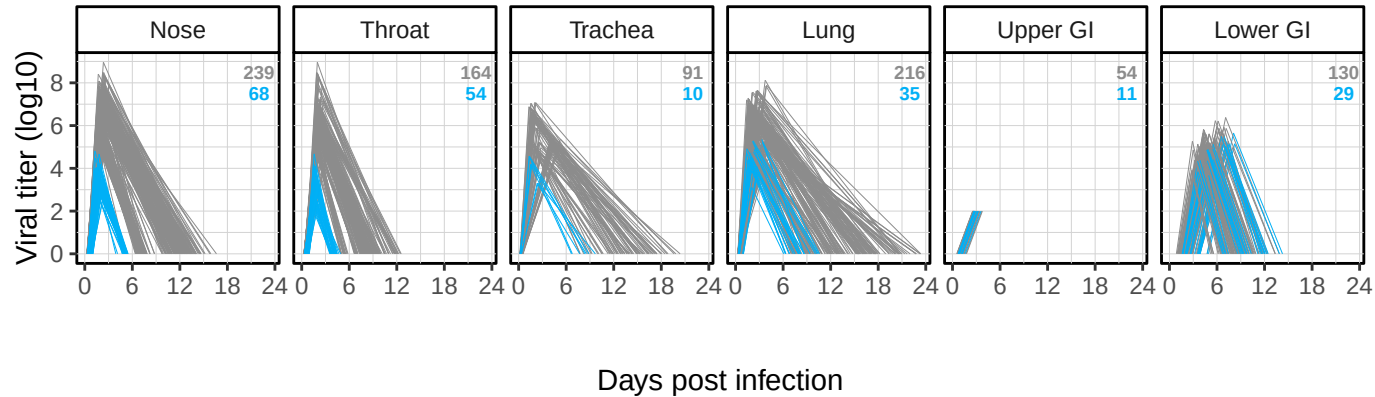


A

Probability of positivity (%)



B

Detection Assay ■ Culture ■ PCR

C

Effects of cofactors on model predictions

Tissue Sampled	Probability of positivity (%)					Time to detectability (days)					Time to peak titer (days)					Peak titer (log10 pfu)				
	sex	species	age	dose	route	sex	species	age	dose	route	sex	species	age	dose	route	sex	species	age	dose	route
Nose	6.6	5.9	3.9	12.7	26.7	0.09	0.09	0.22	0.15	0.34	0.11	0.11	0.38	0.18	0.32	0.67	0.31	0.18	0.65	1.11
Throat	6.4	5.8	3.8	13.1	18	0.09	0.08	0.21	0.14	0.23	0.11	0.11	0.38	0.23	0.48	0.66	0.31	0.18	0.58	1.22
Trachea	2.7	8.9	8	6.7	10.8	0.06	0.09	0.11	0.11	0.26	0.2	0.74	0.37	0.65	1.63	0.12	0.51	0.23	0.69	1.19
Lung	2.4	7.8	7.2	7.9	14.8	0.07	0.11	0.13	0.1	0.28	0.18	0.72	0.38	0.59	1.58	0.11	0.55	0.21	0.7	1.17
Upper GI	4.7	11.9	4.9	12.5	22	0.1	0.32	0.22	0.35	0.67										
Lower GI	5.3	13.5	5.5	11	8.9	0.18	0.57	0.4	0.53	1.42	0.25	0.74	0.67	0.84	2.12	0.5	0.24	0.43	0.76	0.81
Tissue Sampled	Time to undetectability (days)					Duration of infection (days)					AUC									
	sex	species	age	dose	route	sex	species	age	dose	route	sex	species	age	dose	route					
Nose	0.27	0.27	0.68	0.47	1.04	0.35	0.25	0.51	0.55	1.3	0.94	0.89	1.22	2.57	4.61					
Throat	0.27	0.24	0.64	0.39	0.83	0.35	0.22	0.48	0.38	0.8	0.86	0.77	1.06	1.64	3.21					
Trachea	2.15	1.71	1.07	1.36	2.98	2.2	1.64	0.99	1.34	2.86	4.02	5.37	2.11	5.53	9.47					
Lung	2.12	1.83	1.06	1.08	3.39	2.19	1.74	0.97	1.09	3.34	4.57	6.04	2.15	3.35	7.67					
Upper GI																				
Lower GI	0.44	1.08	1.25	1.69	3.36	0.45	0.63	0.94	1.31	2.06	2.43	2	3.42	5.8	7.46					

Cofactor

Relative Effect: 0.0 0.5 1.0

D

